Tianye Song

clairesong24@gmail.com | 412-933-9177 | linkedin.com/in/tianye-song | San Jose, CA

PROFESSIONAL SUMMARY

AI/ML engineer and technical architect with a track record of owning GenAI and ML systems end to end, from system design through production deployment, translating ambiguous business problems into scalable platforms that measurably cut engineering cost and risk. Partners directly with engineering and business leadership to set technical direction and drive enterprise adoption of AI-powered capabilities.

EDUCATION

Carnegie Mellon University – M.S. in Information Systems Management, GPA: 3.75 *12/2022*

Xiamen University – B.Econ. in Finance, GPA: 3.8 (Oxford Winter Exchange, GPA: 4.0) *06/2020*

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, R, TypeScript, JavaScript

Tools & Platforms: PyTorch, Scikit-learn, Spark, Hive, Hadoop, MySQL, MongoDB, Docker, AWS, Claude Code, Copilot

GenAI & Agentic AI: Agentic Code Generation, Multi-Step Agent Workflows, LLM-Powered Tooling

WORK EXPERIENCE

Machine Learning & AI Product Engineer – Synopsys *San Jose, CA | 02/2023 – Present*

Led the AI Code Generation Platform & Developer Tooling

- Architected and built an LLM-powered assistant from the ground up across multiple engineering simulation product lines, each with its own simulation design language, owning the database, backend, frontend, and workflow configuration end to end, and shipping it through both a public web tool and an embedded feature in the product's unified installer.
- Built the LLM harness and multi-step workflow orchestration powering the assistant's ability to answer questions and generate simulation design language, applying reinforcement learning to continuously improve output quality, and developing reusable components that other product teams adopted directly and cutting integration time by 50%.
- Defined the evaluation framework and code-review standards for AI-generated output (GitHub Copilot, Claude Code) spanning correctness, security, and performance, establishing the team's practice for responsible AI-assisted development.

Owned the Cloud ML Platform & Online Estimator

- Engineered Datadog observability instrumentation (traces, logs, and metrics) across the ML platform to detect anomalies in real time, cutting issue identification time by 80%.
- Built a cloud-native ML pipeline (Kubernetes, Docker, Argo Workflows) to orchestrate large-scale simulation workloads, and redesigned CI/CD based on developer feedback, cutting execution time by 50% and improving platform reliability for downstream engineering teams.
- Designed and deployed the Online Estimator prediction service on AWS Lambda and EC2, combining a GBDT model with a PyTorch neural network, and presented the design and results directly to engineering leadership, driving a 30% improvement in prediction accuracy and strengthening confidence in automated engineering estimates.

Data Scientist – Meritco Services Technology Consulting Firm *Shanghai, China | 07/2020 – 07/2021*

- Developed data-driven promotion strategies from customer transaction analysis (Python, SQL) and advised client leadership on CRM targeting, lifting conversion rate by 5% and ROI by 7%.
- Architected a multi-source sales forecasting pipeline and presented the resulting models directly to manufacturing planning stakeholders, improving forecast accuracy by 50% and informing production planning decisions.
- Delivered an XGBoost-based SKU name-matching system to replace manual lookup, accelerating processing time 120x and removing a recurring manual workload for business teams.
- Fine-tuned a pretrained BERT model for NER (TensorFlow) to automatically extract cosmetic product names and features for a Key Opinion Leader product, achieving 87% accuracy and cutting manual processing time by 80%.

ACADEMIC PROJECTS

Movie Recommendation System *01/2022 – 05/2022*

- Created a recommendation system (1M users, 27K movies) using Kafka, MongoDB, and a collaborative filtering model; containerized with Docker for A/B testing and implemented Jenkins for CI and automated retraining.

Speech Recognition System *04/2022 – 05/2022*

- Implemented wav2vec 2.0, k-means clustering, and an RNN + Attention architecture for end-to-end speech-to-text transcription.